

Numerical Methods for DAEs

Another good reference:
Ascher + Petzold
Comp. Meth. for ODEs
and DAEs.

Implicit DAE:

$$F(w(t), w'(t)) = 0$$

Semi-explicit DAE:

$$y' = f(y, z)$$

$$g(y, z) = 0$$

Direct Discretization / State Space Formulation

An example:

$$u'(t) + \eta t v'(t) + (1 + \eta) v(t) = 0$$

$$u(t) + \eta t v = q(t)$$

$$\eta \in \mathbb{R}$$

$q(t)$ given

Exact solution:

$$q'(t) = u'(t) + \eta v(t) + \eta t v'(t)$$

$$q'(t) = u' + \eta t v' + \eta v$$

$\Rightarrow$

$$v(t) = -q'(t)$$

$$u(t) = q(t) + \eta t q'(t)$$

The analysis of this example
seems to be incorrect (i.e.,
direct discretization works fine)
See my other note.

In general it's dangerous to directly
discretize DAEs of index ≥ 1 .

Reformulation can be difficult, so direct
discretization is convenient when index=1.

$$\frac{u_n - u_{n-1}}{h} + (1+\eta)v_n = 0$$

$$y_n^1 + \eta t_n y_n^2 = q(t_n) \quad \checkmark$$

$$\frac{y_n^1 - y_{n-1}^1}{h} + \eta t_n \frac{y_n^2 - y_{n-1}^2}{h} + (1+\eta)y_n^2 = 0$$

Acher & Petzold mistakenly set

$$\begin{bmatrix} 0 & 0 \\ 1 & \eta t_n \end{bmatrix} \frac{\vec{y}_n - \vec{y}_{n-1}}{h} = \begin{bmatrix} 0 \\ \frac{y_n^1 - y_{n-1}^1}{h} + \eta t_n \frac{y_n^2 - y_{n-1}^2}{h} \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ \frac{y_n^1 + \eta t_n y_n^2 - (y_{n-1}^1 + \eta t_n y_{n-1}^2)}{h} \end{bmatrix} \neq \begin{bmatrix} 0 \\ \frac{q(t_n) - q(t_{n-1})}{h} \end{bmatrix}$$

Recommended methods: BDF or RK Collocation
especially Radau

Note that we can even use explicit methods
if the ODEs are non-stiff.

RK Methods

Direct discretization of

$$y' = f(y, z)$$

$$g(y, z) = 0$$

gives

$$\begin{aligned} y_i &= y_n + h \sum_{j=1}^s a_{ij} f(y_j, z_j) \\ g(y_j, z_j) &= 0 \end{aligned} \quad \left. \vphantom{\begin{aligned} y_i &= y_n + h \sum_{j=1}^s a_{ij} f(y_j, z_j) \\ g(y_j, z_j) &= 0 \end{aligned}} \right\} \text{coupled system to solve}$$

$$y_{n+1} = y_n + h \sum_{j=1}^s b_j f(y_j, z_j) \leftarrow \text{explicit}$$

$$g(y_{n+1}, z_{n+1}) = 0 \leftarrow \text{solve for } z_{n+1}.$$

Works well for index-1 problems.

ε -embedding method

Apply RK to SPP

assume A^{-1}
exists

$$y' = f(y, z)$$

$$\varepsilon z' = g(y, z)$$

to get

$$Y_i = y_n + h \sum_{j=1}^s a_{ij} f(Y_j, Z_j)$$

$$\varepsilon Z_i = \varepsilon z_n + h \sum_{j=1}^s a_{ij} g(Y_j, Z_j)$$

$$y_{n+1} = y_n + h \sum_{j=1}^s b_j f(Y_j, Z_j)$$

$$\varepsilon z_{n+1} = \varepsilon z_n + h \sum_{j=1}^s b_j g(Y_j, Z_j)$$

Define

$$Z = \begin{bmatrix} z_1 \\ \vdots \\ z_s \end{bmatrix}$$

$$G = \begin{bmatrix} G(Y_1, Z_1) \\ \vdots \\ G(Y_s, Z_s) \end{bmatrix}$$

$$\mathbf{1} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

$$e_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \quad \text{entry } j$$

$$\Rightarrow \varepsilon(Z - z_n \mathbf{1}) = hAG \Rightarrow \varepsilon A^{-1}(Z - z_n \mathbf{1}) = hG$$

$$\Rightarrow \cancel{\varepsilon} z_{n+1} = \cancel{\varepsilon} z_n + \cancel{\varepsilon} b^T \overset{hG}{\downarrow} A^{-1}(Z - z_n \mathbf{1})$$

Now take $\varepsilon \rightarrow 0$

$$Y_i = Y_n + h \sum_{j=1}^s a_{ij} f(Y_j, Z_j)$$

$$\sum_{j=1}^s a_{ij} g(Y_j, Z_j) = 0$$

$$Y_{n+1} = Y_n + h \sum_{j=1}^s b_j f(Y_j, Z_j)$$

$$Z_{n+1} = Z_n + b^T A^{-1} (Z - Z_n \mathbb{1})$$

ε -embedding
method

$$\Rightarrow Z_{n+1} = (1 - b^T A^{-1} \mathbb{1}) Z_n + b^T A^{-1} Z \quad \text{How accurate is } Z_{n+1}?$$

Recall:

$$R(x) = 1 + x b^T (I - xA)^{-1} \mathbb{1}$$

$$x \neq 0: \quad R(x) = 1 + b^T \left(\frac{1}{x} I - A \right)^{-1} \mathbb{1}$$

$$R(\infty) = 1 - b^T A^{-1} \mathbb{1}$$

Notice that if $R(\infty) = 0$ (i.e. the method is L-stable) then errors in z aren't propagated between steps.

A sufficient for $R(\infty) = 0$ is

$$a_{sj} = b_j \quad j=1, 2, \dots, s \quad (1)$$

Proof: We have $A^T e_s = b$

$$\Rightarrow b^T = e_s^T A \Rightarrow b^T A^{-1} \mathbb{1} = e_s^T A A^{-1} \mathbb{1} = 1.$$

A method that satisfies (1) is called "stiffly accurate".

HW Thm. VI.1.1 (convergence for ε -embedding)

Let the index-1 semi-explicit DAE

$$y' = f(y, z)$$

$$g(y, z) = 0$$

be given (with $g_z \neq 0$).

Apply the ε -emb. method using a RK with order p , stage order q , and A invertible.

$$\text{Then } z_n - z(t_n) = \mathcal{O}(h^r)$$

Where:

a) $r = p$ for stiffly accurate,

b) $r = \min(p, q+1)$ if $-1 \leq R(\infty) < 1$,

c) $r = \min(p-1, q)$ if $R(\infty) = 1$,

d) z diverges if $|R(\infty)| > 1$.

In a way, DAEs are "easier" than stiff ODEs, with this approach.

Approaches for higher-index problems

(See HW section VII.2)

- Index reduction (differentiate until index 1)
- Project onto the constraint manifold at each step.
- Local state space form: find coordinates that exactly represent the manifold locally, and integrate in those coordinates.